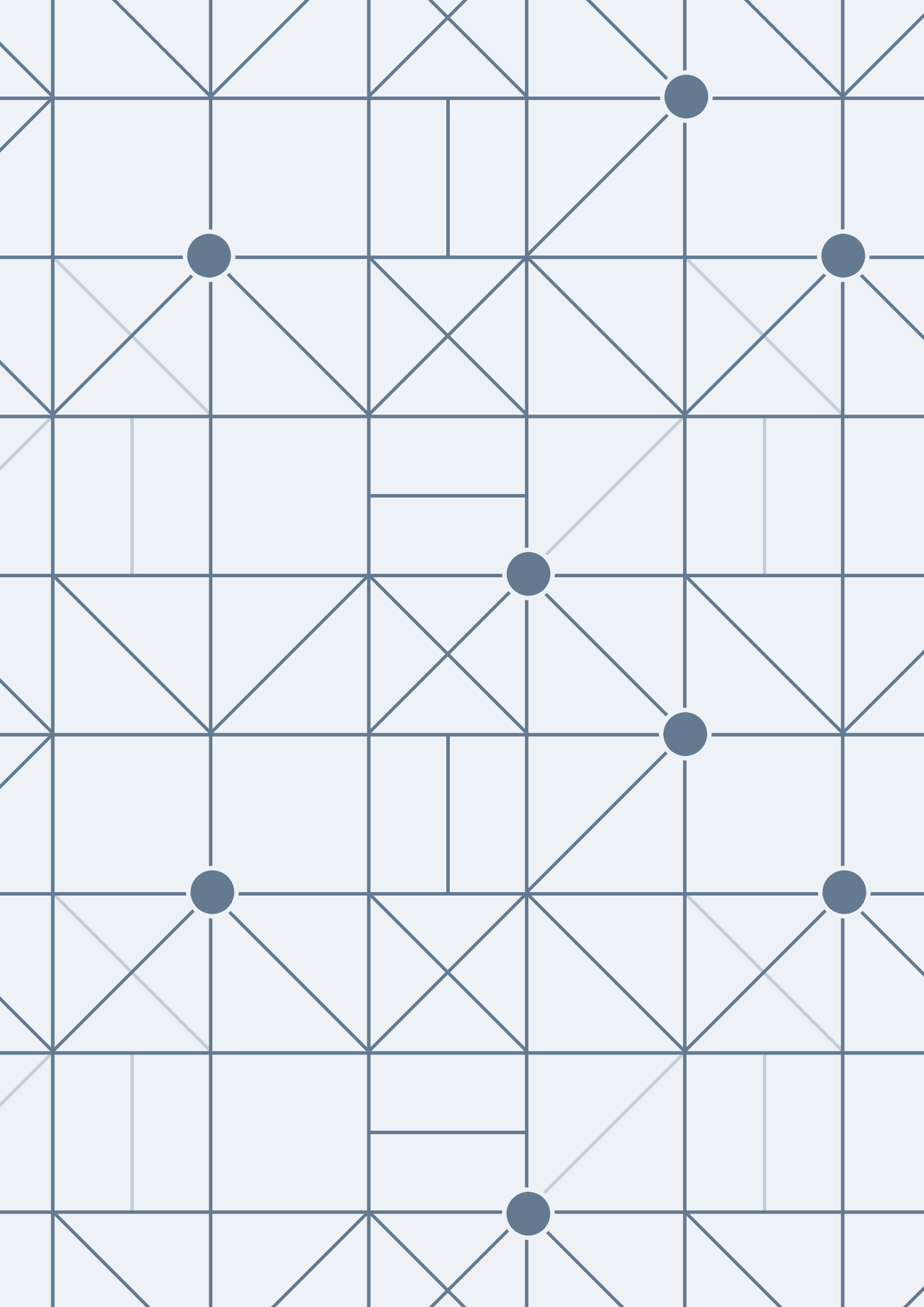




Def 1

Let $\mathcal{H}$ and $\mathcal{H}'$ be Hilbert spaces

$T: \mathcal{H} \rightarrow \mathcal{H}'$ is a linear operator if it is linear and bounded

$\|T\| = \sup_{\|x\|=1} \|Tx\|$ norm of T (operator norm)

$$\|Tx\| = \sqrt{\langle Tx, Tx \rangle} = \sqrt{\langle T^*Tx, x \rangle} = \sqrt{\langle \sum_{j=1}^n \lambda_j^2 e_j, x \rangle} = \sqrt{\sum_{j=1}^n \lambda_j^2 |x_j|^2} = \sqrt{\sum_{j=1}^n \lambda_j^2} \|x\|$$

operator norm

$$\|T\| = \sqrt{\sum_{j=1}^n \lambda_j^2} = \sqrt{\sum_{j=1}^n \lambda_j^2} = \sqrt{\sum_{j=1}^n \lambda_j^2}$$

Let $T: \mathcal{H} \rightarrow \mathcal{H}'$ be a linear operator. Then T is self-adjoint if and only if $T = T^*$.

Let $T: \mathcal{H} \rightarrow \mathcal{H}'$ be a linear operator. Then T is normal if and only if $TT^* = T^*T$.

Let $T: \mathcal{H} \rightarrow \mathcal{H}'$ be a linear operator. Then T is unitary if and only if $T^*T = I$ and $TT^* = I$.

Normal operator

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad T^* = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}$$

for normal operator

Let T be a normal operator. Then T is self-adjoint if and only if $T = T^*$.

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad T^* = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix} \quad T = T^* \Leftrightarrow \begin{cases} a = \bar{a} \\ b = \bar{c} \\ c = \bar{b} \\ d = \bar{d} \end{cases}$$

if T is self-adjoint, then $T = T^*$ and T is normal

A very important example is the Laplacian on a graph

Example:

Let G be a graph with n vertices

$$L = \frac{1}{2} \sum_{i,j \in E} (x_i - x_j)^2$$

Let $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$

$$Lx = \frac{1}{2} \sum_{i,j \in E} (x_i - x_j)^2 = \frac{1}{2} \sum_{i,j \in E} (x_i^2 - 2x_i x_j + x_j^2) = \frac{1}{2} \sum_{i,j \in E} x_i^2 - \sum_{i,j \in E} x_i x_j + \frac{1}{2} \sum_{i,j \in E} x_j^2$$

Let A be the adjacency matrix

$$Lx = \frac{1}{2} \sum_{i,j \in E} x_i^2 - \sum_{i,j \in E} x_i x_j + \frac{1}{2} \sum_{i,j \in E} x_j^2 = \frac{1}{2} \sum_{i,j \in E} x_i^2 - \sum_{i,j \in E} x_i x_j + \frac{1}{2} \sum_{i,j \in E} x_j^2 = \frac{1}{2} \sum_{i,j \in E} x_i^2 - \sum_{i,j \in E} x_i x_j + \frac{1}{2} \sum_{i,j \in E} x_j^2$$

if T is self-adjoint, then $T = T^*$ and T is normal

Example 2: Let T be a linear operator

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad T^* = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}$$

if T is self-adjoint, then $T = T^*$ and T is normal